Project 2 – TMA4300

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#Library imports

```r
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.17
## Current TMB package version is 1.9.18
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more
```

Problem 1

We suppose that the joint density of U, Z up to a normalizing constant is

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

a)

If a joint density is known up to proportionality, the full conditional of one variable is obtained by

$$f_{X|Y}(x|y) \propto f^*_{U,Z}(u,z) \textit{as a function of } x$$

Conditional ditribution of $U|Z = z$

Fix z. The joint density becomes

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

As a function of u, this is constant on the interval $(0, e^{-z^2/2})$.

The normalizing constant is

$$\int_0^{e^{-z^2/2}} 1\, du = e^{-z^2/2}.$$

Therefore,

$$f_{U|Z}(u \mid z) = \frac{1}{e^{-z^2/2}} = e^{z^2/2}, \quad 0 < u < e^{-z^2/2}.$$

Hence,

$$U \mid Z = z \sim \text{Unif}\big(0, e^{-z^2/2}\big).$$

.

Conditional ditribution of $Z|U = u$

Fix u. The support condition is $0 < u < e^{-z^2/2}$.

Solving for z:

$$u < e^{-z^2/2} \quad \Longleftrightarrow \quad \log u < -\frac{z^2}{2}$$

$$-2\log u > z^2$$

$$|z| < \sqrt{-2\log u}$$

Define

$$a(u) = \sqrt{-2\log u}$$

Therefore, the support of the conditional distribution is

$$z \in (-a(u), a(u)).$$

Within this interval the joint density is constant and equal to 1. The normalizing constant is therefore the length of the interval:

$$\int_{-a(u)}^{a(u)} 1\, dz = 2a(u).$$

Thus, the full conditional density of $Z \mid U = u$ is

$$f_{Z|U}(z \mid u) = \frac{1}{2a(u)}, \quad |z| < a(u),$$

where

$$a(u) = \sqrt{-2\log u}.$$

Thus,

$$Z \mid U = u \sim \text{Unif}\big(-a(u), a(u)\big).$$

Gibbs Sampler

Given an initial valid point $(U^{(0)}, Z^{(0)})$ satisfying $0 < U^{(0)} < e^{-(Z^{(0)})^2/2}$, iterate for $t = 1, \ldots, n$:

1. Sample
$$U^{(t)} \sim \text{Unif}\left(0, e^{-(Z^{(t-1)})^2/2}\right).$$

2. Define
$$a^{(t)} = \sqrt{-2\log U^{(t)}}.$$

3. Sample
$$Z^{(t)} \sim \text{Unif}\left(-a^{(t)},\, a^{(t)}\right).$$

This iterative scheme defines a Markov chain whose stationary distribution is the joint density $f_{U,Z}(u, z)$.The marginal stationary distribution of Z is $\mathcal{N}(0, 1)$.

```r
gibbs_normal <- function(n, u_init = 0.5, z_init = 0) {

  U <- numeric(n)
  Z <- numeric(n)

  U[1] <- u_init
  Z[1] <- z_init

  for (t in 2:n) {

    # Sample U | Z
    U[t] <- runif(1, 0, exp(-(Z[t-1]^2)/2))

    # Compute truncation bound
    a <- sqrt(-2 * log(U[t]))

    # Sample Z | U
    Z[t] <- runif(1, -a, a)
  }

  Z  # return only Z since that is the target
}
```

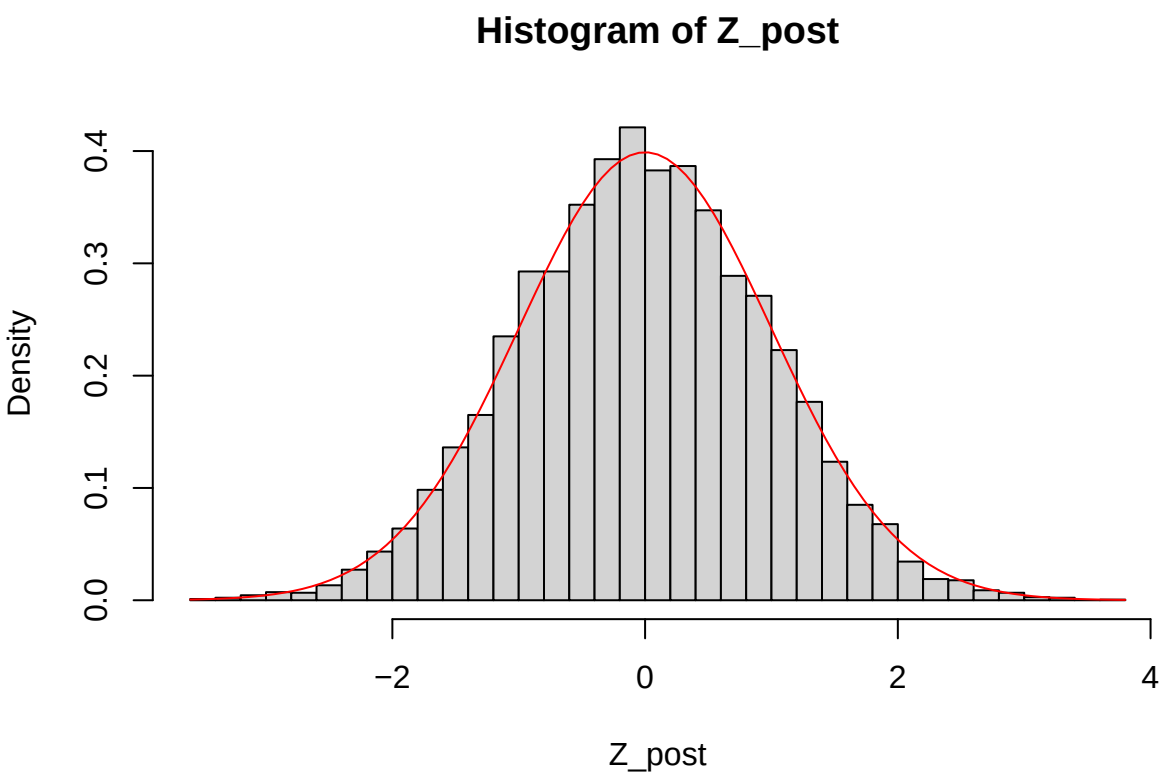
```r
# Run Gibbs sampler
set.seed(123)
Z <- gibbs_normal(10000)

# Burn-in removal
Z_post <- Z[1001:10000]

# Diagnostics
mean(Z_post)
```

```
## [1] -0.01857042
```

```r
var(Z_post)
```

```
## [1] 0.9699554
```

```r
hist(Z_post, probability = TRUE, breaks = 40)
curve(dnorm(x), col="red", add=TRUE)
```



The posterior sample mean is approximately

$$\hat{\mu} \approx -0.019,$$

and the empirical variance is

$$\hat{\sigma}^2 \approx 0.97.$$

Both values are close to the theoretical mean and variance,

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1,$$

of a standard normal distribution.

The histogram aligns well with the standard normal density curve, confirming that the Gibbs sampler correctly targets

$$Z \sim \mathcal{N}(0, 1).$$

b)

We determine the marginal distribution of Z by integrating out U.

Recall that the joint density (up to normalization) is

$$f^*_{U,Z}(u, z) = \begin{cases} 1, & 0 < u < e^{-z^2/2}, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal density of Z is therefore

$$f_Z(z) = \int_0^{e^{-z^2/2}} 1 \, du = e^{-z^2/2}.$$

Thus,

$$f_Z(z) \propto e^{-z^2/2}.$$

Since

$$\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi},$$

the normalized density is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Therefore,

$$Z \sim \mathcal{N}(0,1).$$

Hence, discarding the U-values from the Gibbs sampler yields a sample from a standard normal distribution.

c)

Correlation between successive Gibbs samples Z_{n-1} and Z_n}

From the Gibbs sampler we have

$$Z_n \mid U_n \sim \mathrm{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

This conditional distribution is symmetric around zero. Therefore,

$$\mathbb{E}[Z_n \mid U_n] = 0.$$

By the law of total expectation,

$$\mathbb{E}[Z_n] = \mathbb{E}[\mathbb{E}[Z_n \mid U_n]] = \mathbb{E}[0] = 0.$$

Since the Gibbs sampler has stationary distribution $Z_n \sim \mathcal{N}(0,1)$, it follows that

$$\mathbb{E}[Z_{n-1}] = \mathbb{E}[Z_n] = 0.$$

We now compute the covariance. By definition,

$$\mathrm{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n] - \mathbb{E}[Z_{n-1}]\mathbb{E}[Z_n].$$

Because both expectations are zero, this reduces to

$$\mathrm{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n].$$

Using the tower property (law of iterated expectation),

$$\mathbb{E}[Z_{n-1}Z_n] = \mathbb{E}[\mathbb{E}[Z_{n-1}Z_n \mid U_n]]\,.$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent. Therefore,

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = Z_{n-1}\,\mathbb{E}[Z_n \mid U_n].$$

Since $\mathbb{E}[Z_n \mid U_n] = 0$, we obtain

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = 0.$$

Taking expectations again,

$$\mathbb{E}[Z_{n-1}Z_n] = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}, Z_n) = 0.$$

Successive Gibbs samples Z_{n-1} and Z_n are uncorrelated in stationarity. However, they are not independent, since both depend on the intermediate variable U_n.

Correlation between successive second moments Z_{n-1}^2 and Z_n^2}

Recall that

$$Z_n \mid U_n \sim \mathrm{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

For a symmetric uniform distribution on $(-a, a)$ we have

$$\mathbb{E}[X^2] = \frac{a^2}{3}.$$

Therefore,

$$\mathbb{E}[Z_n^2 \mid U_n] = \frac{a(U_n)^2}{3} = \frac{-2\log U_n}{3}.$$

Unlike the first moment case, this conditional second moment depends explicitly on U_n.

We now compute the covariance using the law of total covariance:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathbb{E}\big[\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n)\big] + \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big)\,.$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent, and therefore

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n) = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big)\,.$$

From the previous calculation,

$$\mathbb{E}[Z_{n-1}^2 \mid U_n] = \mathbb{E}[Z_n^2 \mid U_n] = \frac{-2\log U_n}{3}.$$

Therefore,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}\left(\frac{-2\log U_n}{3}\right).$$

To evaluate this variance, we determine the marginal distribution of U_n in stationarity.
It can be shown that

$$T = -2\log U_n \sim \chi_3^2.$$

Hence,

$$\mathbb{E}[T] = 3, \qquad \mathrm{Var}(T) = 6.$$

It follows that

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}(T/3) = \frac{6}{9} = \frac{2}{3}.$$

Successive second moments are positively correlated:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \frac{2}{3} \neq 0.$$

Thus, although successive Gibbs samples Z_{n-1} and Z_n are uncorrelated, their squared values exhibit positive correlation due to the shared dependence on the intermediate variable U_n.

Problem 2

a)

```r
load(url("https://www.math.ntnu.no/emner/TMA4300/2026v/random-intercept-data.RData"))

#str(data)
#head(data)
```

```r
mod <- glmmTMB(y ~ (1|i), data=data)
summary(mod)
```

```
##  Family: gaussian  ( identity )
## Formula:          y ~ (1 | i)
## Data: data
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     280.3     288.1    -137.2     274.3        97
##
## Random effects:
##
## Conditional model:
##  Groups   Name        Variance Std.Dev.
##  i        (Intercept) 0.2301   0.4797
##  Residual             0.7560   0.8695
## Number of obs: 100, groups:  i, 20
##
## Dispersion estimate for gaussian family (sigma^2): 0.756
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  10.1888     0.1381   73.79   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The maximum likelihood estimate of the overall mean is $\hat{\mu} = 10.1888$. The estimated variance of the random intercepts is $\widehat{\mathrm{Var}}(\gamma_i) = 0.2301$, corresponding to $1/\nu$, and the estimated residual variance is $\widehat{\mathrm{Var}}(\epsilon_{ij}) = 0.7560$, corresponding to $1/\tau$. These estimates will later be used to check that the Gibbs sampler implementation produces reasonable results.

b) Full conditional distributions

From the hierarchical model

$$y_{ij} \mid \mu, \gamma_i, \tau \sim N(\mu + \gamma_i, \tau^{-1}),$$

$$\gamma_i \mid \nu \sim N(0, \nu^{-1}),$$

with priors

$$\pi(\mu) \propto 1, \qquad \tau \sim \text{Gamma}(\alpha, \alpha), \qquad \nu \sim \text{Gamma}(\alpha, \alpha),$$

we derive the following full conditional distributions.

Let $N = \sum_{i=1}^{m} n_i$.

Full conditional of μ Combining the likelihood with the flat prior for μ, we obtain

$$\mu \mid \gamma, \tau, y \sim N(\bar{y}_\gamma, (N\tau)^{-1}),$$

where

$$\bar{y}_\gamma = \frac{1}{N} \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \gamma_i).$$

Full conditional of γ_i Using the likelihood contribution from cluster i together with the prior $\gamma_i \sim N(0, \nu^{-1})$, we obtain

$$\gamma_i \mid \mu, \tau, \nu, y \sim N(m_i, (\nu + n_i\tau)^{-1}),$$

where

$$m_i = \frac{\tau \sum_{j=1}^{n_i} (y_{ij} - \mu)}{\nu + n_i\tau}.$$

Full conditional of τ Let

$$S = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \mu - \gamma_i)^2.$$

Combining the Gaussian likelihood with the prior $\tau \sim \text{Gamma}(\alpha, \alpha)$ gives

$$\tau \mid \mu, \gamma, \nu, y \sim \text{Gamma}\left(\alpha + \frac{N}{2}, \alpha + \frac{S}{2}\right).$$

Full conditional of ν Using the prior for γ_i and $\nu \sim \text{Gamma}(\alpha, \alpha)$, we obtain

$$\nu \mid \mu, \gamma, \tau, y \sim \text{Gamma}\left(\alpha + \frac{m}{2}, \alpha + \frac{1}{2} \sum_{i=1}^{m} \gamma_i^2\right).$$

c) Implementation of a Gibbs sampler

```
gibbs_sampler <- function(data, n_iter = 20000, alpha = 0.0001){

y <- data$y
i <- data$i
```

```r
groups <- unique(i)
m <- length(groups)
N <- length(y)

n_i <- table(i)

mu <- mean(y)
gamma <- rep(0,m)
tau <- 1
nu <- 1

samples <- matrix(NA, n_iter, m+3)

for (iter in 1:n_iter){

# update mu
y_gamma <- y - gamma[i]
mu <- rnorm(1, mean(y_gamma), sqrt(1/(N*tau)))

# update gamma_i
for (k in 1:m){

y_k <- y[i==groups[k]]
n_k <- length(y_k)

var_gamma <- 1/(nu + n_k*tau)
mean_gamma <- (tau*sum(y_k-mu))/(nu+n_k*tau)

gamma[k] <- rnorm(1, mean_gamma, sqrt(var_gamma))
}

# update tau
res <- y - mu - gamma[i]
tau <- rgamma(1, alpha + N/2, alpha + sum(res^2)/2)

# update nu
nu <- rgamma(1, alpha + m/2, alpha + sum(gamma^2)/2)

samples[iter,] <- c(mu, gamma, 1/tau, 1/nu)
}

return(samples)
}
```

```r
samples <- gibbs_sampler(data, 20000)
colMeans(samples)
```

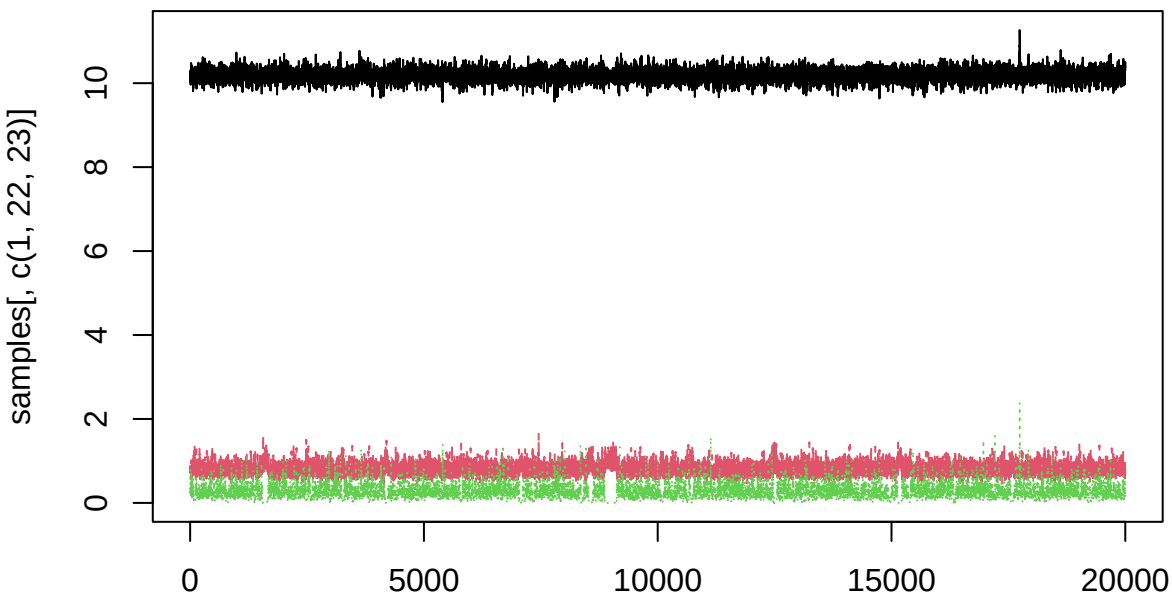
```
##  [1] 10.186949480 -0.228328265 -0.241998135 -0.308130141  0.445724338
##  [6]  0.001276209 -0.195878016  0.081798904  0.209191417  0.310980433
## [11]  0.052458426  0.112042491 -0.069377360 -0.319505738 -0.513020222
## [16]  0.656524132 -0.441780123 -0.323819534  0.752023533  0.063353793
## [21] -0.013827620  0.802720740  0.246525959
```

The Gibbs sampler was run for 20000 iterations to sample from the joint posterior distribution of the model

parameters. The posterior mean of the global intercept is approximately $\mu \approx 10.19$, which is very close to the maximum likelihood estimate obtained in part (a). The posterior mean of the residual variance is approximately $1/\tau \approx 0.80$ and the posterior mean of the random intercept variance is approximately $1/\nu \approx 0.25$. These values are close to the corresponding maximum likelihood estimates (0.756 and 0.2301), suggesting that the Gibbs sampler produces reasonable estimates and appears to converge to the correct posterior distribution.

d)

```
matplot(samples[,c(1,22,23)], type="l")
```



The trace plots for the parameters appear stable and fluctuate around constant levels, indicating that the Gibbs sampler mixes reasonably well and that the Markov chains have reached a stationary distribution. The chain for the global intercept μ is centered around approximately 10.2, while the residual variance $1/\tau$ fluctuates around about 0.8 and the random intercept variance $1/\nu$ around about 0.25. These values are consistent with the maximum likelihood estimates obtained in part (a). The trace plot for $1/\nu$ shows slightly larger variability and occasional spikes compared to the other parameters, which is expected for variance parameters with weakly informative priors, but overall the chain still appears stable.